24/04/2025

CODE-A

Aakash

Medical | IIT-JEE | Foundations

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AIM – 720

(*Advanced* INTENSIVE *Mastery for* 720)

MM : 720

CST-10

Time : 180 Mins.

Complete Syllabus of NEET

Instructions:

(i) Duration of Test is 3 hrs.

(ii) The Test consists of 180 questions. The maximum marks are 720.

(iii) There are three parts in the question paper consisting of Physics, Chemistry and Biology having 45 questions each in Physics and Chemistry and 90 questions in Biology).

(iv) Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.

(v) Use blue/black ballpoint pen only to darken the appropriate circle.

(vi) Mark should be dark and completely fill the circle.

(vii) Dark only one circle for each entry.

(viii) Dark the circle in the space provided only.

(ix) Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. The temperature of a metallic wire is increased. The Young's modulus of elasticity of the wire

(1) Will increase

(2) Will decrease

(3) Will remains same

(4) Either (1) or (3)

2. Consider the given statements
Statement A : Acceleration of a particle, executing SHM, at mean position is zero.
Statement B : The velocity of oscillating bob in a simple harmonic motion is maximum at the extreme position.
Statement C: Total energy of a particle in SHM remains constant
Which of the following statement(s) is/are correct?

(1) A and B only

(2) B and C only

(3) A and C only

(4) A, B and C

3. A string fixed at both ends oscillates in 4 segments. If length of string is 10 m and velocity of wave is 20 m/s, then frequency is

(1) 4 Hz

(2) 5 Hz

(3) 10 Hz

(4) $\frac{1}{4}$ Hz

4. The first diffraction minima due to a single slit diffraction is at θ = 30° for a light of wavelength 4000 Å. The width of the slit is

(1) 8 μm

(2) 80 μm

(3) 0.8 μm

(4) 800 μm

5. A capacitor of capacitance C_0 is connected across an AC source of voltage V, given by $V = V_0 \sin \omega t$. The displacement current between the plates of the capacitor, would be given by

(1) $I_d = V_0 \omega c_0 \sin \omega t$

(2) $I_d = V_0 \omega c_0 \cos \omega t$

(3) $I_d = \frac{V_0}{\omega c_0} \sin \omega t$

(4) $I_d = \frac{V_0}{\omega c_0} \cos \omega t$

6. Consider the following statements about diamagnetic material and choose the correct option.
a. An atom generally has zero magnetic moment in absence of external magnetic field.
b. Diamagnetism is universal property
c. Magnetic susceptibility decreases with increase in temperature.
d. Superconductors are diamagnetic.

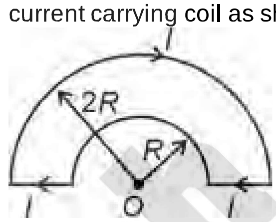
(1) Only a, b and d are correct

(2) Only a, b and c are correct

(3) Only b, c and d are correct

(4) All a, b, c, and d are correct

7. The magnitude of magnetic field induction at the centre O of current carrying coil as shown is



(1) $\frac{\mu_0 I}{2R}$

(2) $\frac{\mu_0 I}{8R}$

(3) $\frac{\mu_0 I}{4R}$

(4) $\frac{\mu_0 I}{16R}$

8. The time period of a charged particle of mass m having charge q which is projected in uniform transverse magnetic field $\vec{B}$ with velocity $(\vec{v})$ is independent of

(1) Charge (q)

(2) Magnetic field $\left(\vec{B}\right)$

(3) Mass of particle (m)

(4) Velocity of particle $\left(\vec{v}\right)$

9. If power factor is $\frac{1}{3}$ in series RC circuit for R = 100 Ω, then C is (f = 50 Hz)

(1) $\frac{25}{\pi}$ μF

(2) $\frac{15}{\pi}$ μF

(3) $\frac{15\sqrt{2}}{\pi}$ μF

(4) $\frac{25\sqrt{2}}{\pi}$ μF

10. Core of transformer is made up of

(1) Soft iron

(2) Steel

(3) Copper

(4) Glass

11. An inductor having inductance 4 mH carrying current I given by $I = (2t + 3t^2)$ A, where t is time in s. The magnitude of emf induced in the coil at $t = 2$ s will be

(1) 24 mV

(2) 48 mV

(3) 56 mV

(4) 12 mV

12. The magnitude of slope of magnetic flux versus time graph is equal to

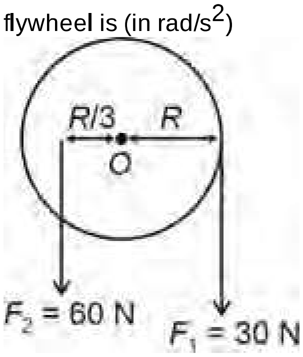
(1) Charge

(2) Energy

(3) Induced emf

(4) Induced charge

13. A flywheel disc of mass 30 kg and radius 20 cm is mounted on a horizontal axle. If forces F_1 and F_2 are applied as shown in figure below. Then angular acceleration of the flywheel is (in rad/s^2)



(1) $\frac{10}{3}$

(2) $\frac{20}{3}$

(3) 10

(4) 20

14. Given below are two statements:
Statement A: In a couple, total external force is zero but total torque is non-zero
Statement B: The total torque on a system depends on the position of the origin, if the net force on system is zero.
In the light of the above statements, choose the most appropriate answer from the options given below.

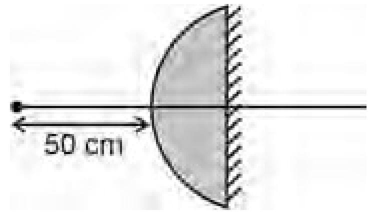
(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Statement A is correct but statement B is incorrect

(4) Statement A is incorrect but statement B is correct

15. A plano convex lens of radius of curvature 20 cm and material having refractive index 1.4 silvered on its flat face as shown in the figure. A point object is located at a distance 50 cm from the surface of lens. The final image will be formed at a distance



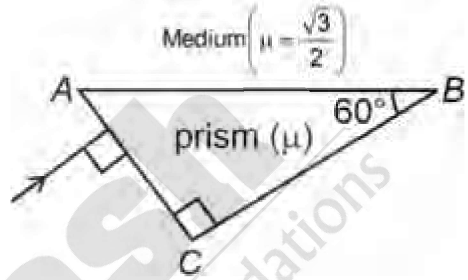
(1) 25 cm from the lens

(2) 50 cm from the lens

(3) At infinity

(4) 100 cm from the lens

16. A ray of light falls normally on face AC of a right angled prism as shown below. If it is incident at critical angle on face AB, then refractive index of the prism is



(1) $\frac{1}{2}$

(2) $\frac{1}{\sqrt{2}}$

(3) $\sqrt{2}$

(4) $\sqrt{3}$

17. The minimum work done to raise a mass m from the surface of the earth to a height 4R is (Where R is radius of earth and g is acceleration due to gravity on surface of earth)

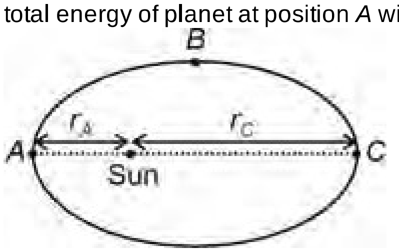
(1) $\frac{3}{4}mgR$

(2) $-\frac{3}{4}mgR$

(3) $\frac{4}{5}mgR$

(4) $-\frac{4}{5}mgR$

18. A planet is revolving around the sun in an elliptical orbit as shown below. Its kinetic energy and potential energy at position C are K_C and V_C respectively. If $r_C = 3r_A$, then total energy of planet at position A will be equal to



(1) $K_C + 3V_C$

(2) $K_C + V_C$

(3) $3K_C + 3V_C$

(4) $\frac{K_C}{3} + 3V_C$

19. The hydrogen atoms are excited to the stationary state designated by the principal quantum number n = 5. The number of spectral lines emitted by these atoms, according to Bohr's theory, will be

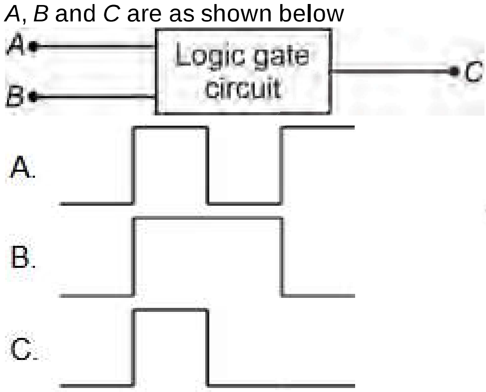
(1) 6

(2) 10

(3) 5

(4) 8

20. The following figure shows a logic gate circuit with two inputs A and B and the output C. The voltage wave forms of A, B and C are as shown below



The logic gate is

(1) AND gate

(2) OR gate

(3) NAND gate

(4) NOR gate

21. Given below are two statements

Statement A: Nuclear force between nucleons is always attractive in nature.

Statement B: The nuclear force does not depend on the electric charge.

In the light of above statements choose the correct option among the following

(1) Statement A is correct but statement B is incorrect

(2) Statement A is incorrect but statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

22. If the zero mark on the vernier scale lies towards the left side of the zero mark of the main scale, when the jaws are closed, then what will be the zero error?

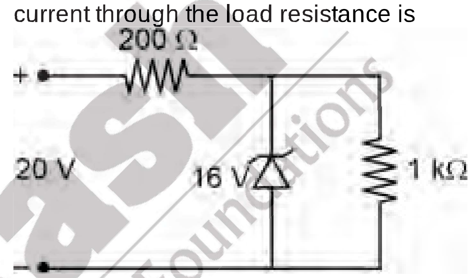
(1) Positive zero error

(2) Negative zero error

(3) Zero error does not exist

(4) Depends on main scale reading

23. A Zener diode, having breakdown voltage equal to 16 V is used in a voltage regulator circuit shown in figure. The current through the load resistance is



(1) 8 mA

(2) 16 mA

(3) 4 mA

(4) Zero

24. The side of a cube is (2.00 ± 0.010) cm. The surface area of the cube is given as

(1) (24.0 ± 0.06) cm^2

(2) (24.0 ± 0.08) cm^2

(3) (24.0 ± 0.09) cm^2

(4) (24.0 ± 0.24) cm^2

25. The magnetic field has the dimensions of

(1) Force

(2) $\frac{Surface\ tension}{Current}$

(3) Surface tension

(4) Work

26. An athlete runs exactly once around a circular track of length 500 m. The runner's displacement in the race is

(1) 100 m

(2) 1000π m

(3) 200 m

(4) Zero

27. An automobile travelling at 40 m/s, can be stopped in time 5 s after the brakes are applied. The time in which the automobile can be stopped when it is travelling at 80 m/s with the same braking force is

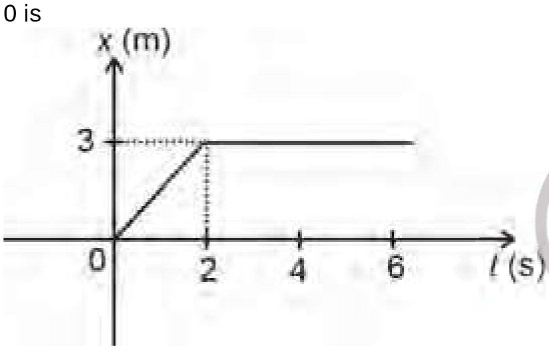
(1) 5 s

(2) 10 s

(3) 15 s

(4) Zero

28. The figure represents the position-time graph of a body of mass 4 kg initially at rest. Impulse imparted to the body at t = 0 is



(1) 6 N s

(2) 3 N s

(3) Zero

(4) 4 N s

29. If a particle is moving in a curved path, it must have a

(1) Component of acceleration parallel to velocity

(2) Constant speed

(3) Constant acceleration

(4) Component of acceleration perpendicular to velocity

30. Displacement x (in meters) of a body of mass 2 kg as a function of time t, on a horizontal smooth surface is given as $x = 4t^2 + 4$. The work done in the first second is

(1) 144 J

(2) 128 J

(3) 168 J

(4) 64 J

31. Given below are two statements: one is labelled as assertion (A) and the other is labelled as reason (R)

Assertion (A) : When non-conservative forces like friction are involved, energy may not be conserved

Reason (R) : In elastic collision, kinetic energy of system before collision is equal to kinetic energy after collision.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

32. Match column-I with column-II with regards to capacitor and choose the correct option.

	Column-I		Column-II
a.	Energy stored in a charged capcitor	(i)	Acts like an open circuit
b.	Fully charged capacitor in DC circuit	(ii)	Associated with the electric field between the plates of capacitor
c.	Dielectric inserted while battery connected	(iii)	Capacitance increases, voltage decreases
d.	Dielectric inserted after battery is removed	(iv)	Capacitance increases, charge increases

(1) a(i), b(ii), c(iii), d(iv)

(2) a(i), b(ii), c(iv), d(iii)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(ii), b(i), c(iv), d(iii)

33. Two point charges $+q$ and $-q$ are placed at $(-a, 0)$ and $(a, 0)$ respectively on a cartesian coordinate system. Then

(1) Electric potential is non zero at every point on y-axis

(2) Electric potential is non zero at every point on x-axis

(3) Electric field is along $+x$ axis at all points on y-axis

(4) Electric field is along $-x$ axis at all points on x-axis

34. Two point charges +2 C and –2 C attract each other with a force of 12 N. If –4 C charge is given to each of these charges, the force will be

(1) 36 N, attractive

(2) 36 N, repulsive

(3) 12 N, repulsive

(4) 18 N, attractive

35. Electric field of an infinitely long uniform charged straight wire having linear charge density λ is proportional to (r is perpendicular distance from the wire)

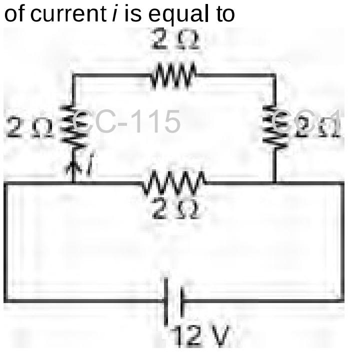
(1) r

(2) $\frac{1}{r^2}$

(3) $\frac{1}{r^3}$

(4) $\frac{1}{r}$

36. A network of resistors is connected with a battery. The value of current i is equal to



(1) 1 A

(2) 2 A

(3) $\frac{2}{3}$ A

(4) $\frac{4}{3}$ A

37. If a copper wire is recasted to make its cross section double, then percentage change in its resistance will be

(1) 25%

(2) 50%

(3) 100%

(4) 75%

38. When a body having density equals to $\frac{1}{3}$ density of water is dipped into water, then fraction of volume outside the water is

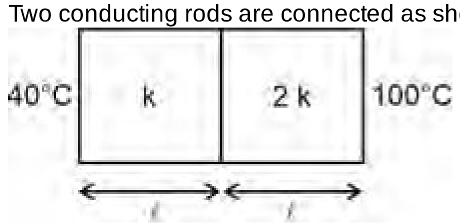
(1) $\frac{1}{2}$

(2) $\frac{1}{3}$

(3) $\frac{2}{3}$

(4) 1

39. Two conducting rods are connected as shown



The rods have same length and same cross-section area. In the steady state the temperature of the interface will be (in °C)

(1) 70°C

(2) 60°C

(3) 80°C

(4) 90°C

40. An ideal gas initially at 0°C temperature is suddenly compressed to one eighth of its volume. If the ratio of specific heat at constant pressure to that at constant volume is $\frac{4}{3}$, then change in temperature due to the thermodynamic process will be

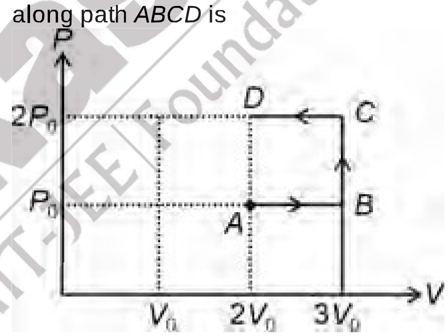
(1) 546 K

(2) 273 K

(3) 173 K

(4) 846 K

41. Consider following *PV*-diagram. Work done by an ideal gas along path *ABCD* is



(1) P_0V_0

(2) $-P_0V_0$

(3) $2P_0V_0$

(4) $4P_0V_0$

42. 1 mole of monoatomic gas is mixed with 2 moles of diatomic gas. Then ratio of specific heat of mixture at constant pressure to specific heat at constant volume $\left(\frac{C_P}{C_V}\right)$ will be

(1) $\frac{19}{13}$

(2) $\frac{20}{13}$

(3) $\frac{13}{20}$

(4) $\frac{13}{19}$

43. A particle is projected with an initial velocity of $20\sqrt{2}\ \mathrm{m\,s^{-1}}$ making an angle 45° with the horizontal. Based on this information match the columns and tick the correct option (g = 10 m/s^2)

	Column-I		**Column-II**
a.	Maximum height reached by the particle (in m)	(i)	15
b.	Time of flight of the particle (in s)	(ii)	80
c.	Horizontal range acquired by the particle (in m)	(iii)	20
d.	Height of the particle at t = 3 s (in m)	(iv)	4

(1) a(i), b(iii), c(ii), d(iv)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(iii), b(ii), c(iv), d(i)

(4) a(ii), b(iv), c(iii), d(i)

44. If alpha particle, deuterium and beta particle all carries same momentum, which has the longest de Broglie wavelength?

(1) Alpha particle

(2) Deuterium

(3) Beta particle

(4) All have same wavelength

45. The pressure at the bottom of a tank containing a liquid does not depend on

(1) Acceleration due to gravity

(2) Height of liquid column

(3) Area of the bottom surface

(4) Density of the liquid

CHEMISTRY

46. Consider the following statements about the product (P) of the given reaction

$$HC \equiv CH \xrightarrow[dil.\,\mathrm{H_2}SO_4 \\ 333\,K]{HgSO_4} \left(P\right)$$

(a) (P) when heated with dilute NaOH, gives but-2-enal as major product.
(b) It reduces Fehling's reagent.
(c) It gives positive iodoform test.
The correct statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

47. Given below are the two statements
Statement I: Insulin is an example of Fibrous protein.
Statement II: Insulin contains 51 amino acid.
In the light of above statements, choose the correct answer

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

48. If the threshold frequency (ν_0) for a metal is 2 × 10^{15} s^{-1} then the kinetic energy of the electron emitted when radiation of frequency 1 × 10^{16} s^{-1} hits the metal will be (h = 6.6 × 10^{-34} J s)

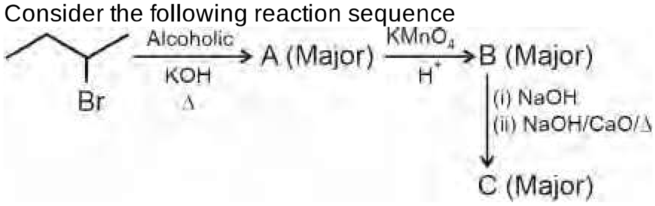
(1) 5.28 × 10^{-18} J

(2) 3.14 × 10^{-19} J

(3) 5.28 × 10^{-20} J

(4) 3.14 × 10^{-17} J

49. Consider the following reaction sequence



Major product (C) is

(1) C_2H_6

(2) C_2H_4

(3) C_2H_2

(4) CH_4

50. Consider the following statements
(a) A unit formed by the attachment of a base to 1′ position of sugar is known as nucleoside.
(b) In DNA molecule, adenine forms hydrogen bond with guanine.
(c) Nucleotides are joined together by phosphodiester linkage between 5′ and 3′ carbon atoms of pentose sugar.
The correct statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

51. The maximum number of emission lines obtained when the excited electron of H atoms in n = 6 drops to the ground state level is

(1) 10

(2) 12

(3) 15

(4) 18

52. The solubility of AgCl(s) in 0.1 M NaCl solution will be (given: K_{sp} (AgCl) = 1.6×10^{-10})

(1) 1.3×10^{-5} M

(2) 3.2×10^{-9} M

(3) 1.6×10^{-11} M

(4) 1.6×10^{-9} M

53. pH of 0.4 M aqueous solution of sodium benzoate at 25°C is (pK_a of benzoic acid = 4.2)

(1) 8.9

(2) 10.4

(3) 11.2

(4) 8.1

54. A solution of $CuSO_4$ is electrolysed for 90 minutes with a current of 4.825 amperes. The mass of copper deposited at the cathode is
(Atomic mass of copper = 63.5 u)

(1) 12.4 g

(2) 17.2 g

(3) 8.6 g

(4) 6.2 g

55. The conductivity of 0.1 M solution of KCl at 298 K is 0.0248 S cm^{-1}, the molar conductivity of the given solution will be

(1) 213 S cm^2 mol^{-1}

(2) 62 S cm^2 mol^{-1}

(3) 248 S cm^2 mol^{-1}

(4) 124 S cm^2 mol^{-1}

56. The value of electron gain enthalpy of Na^+ is (if first ionisation enthalpy of Na = 5.1 eV)

(1) +14.2 eV

(2) +10.2 eV

(3) –5.1 eV

(4) –10.2 eV

57. With which of the following electronic configurations, an atom has the lowest ionisation enthalpy?

(1) $1s^2\ 2s^2\ 2p^6$

(2) $1s^2\ 2s^2\ 2p^5$

(3) $1s^2\ 2s^2\ 2p^4$

(4) $1s^2\ 2s^2\ 2p^3$

58. XeF_2 on complete hydrolysis gives

(1) Xe and HOF

(2) Xe, HF and O_2

(3) XeO_3 and HF

(4) Xe, HOF and O_2

59. Which of the following compounds does not exist?

(1) PF_5

(2) $SbCl_5$

(3) NCl_5

(4) AsF_5

60. Among the following, the correct order of acidic strength is

(1) $HClO_3 < HClO_4 < HClO_2 < HClO$

(2) $HClO_4 < HClO_2 < HClO < HClO_3$

(3) $HClO < HClO_2 < HClO_3 < HClO_4$

(4) $HClO_2 < HClO < HClO_3 < HClO_4$

61. Which of the following will evolve CO_2 on reaction with $NaHCO_3$?
(a) 2,4,6-Trinitrophenol
(b) Benzoic acid
(c) Phenol

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

62. When glycerol is heated with excess of HI, then the major product formed is

(1) Allyl iodide

(2) Propene

(3) 1-iodopropane

(4) 2-iodopropane

63. How many isomers (including stereoisomers) of molecular formula $C_4H_{10}O$ will be possible?

(1) 8

(2) 9

(3) 7

(4) 10

64. Consider the following statements about resonance
(i) Resonating structures are hypothetical and do not represent real molecule.
(ii) The energy of actual structure of the molecule (resonance hybrid) is lower than that of any canonical structures.
(iii) The resonance structures have the same position of nuclei and the same number of unpaired electrons.
The correct statement(s) is/are

(1) (i) and (ii) only

(2) (iii) only

(3) (i) and (iii) only

(4) (i), (ii) and (iii)

65. Match the compounds given in list I with their respective types given in list II

List I		**List II**
a. Pyridine	*(i)*	*Heterocyclic non aromatic compound*
b. Tetrahydrofuran	*(ii)*	*Benzenoid aromatic compound*
c. Tropone	*(iii)*	*Non-benzenoid aromatic compound*
d. Naphthalene	*(iv)*	*Heterocyclic aromatic compound.*

Choose the most appropriate match from the options given below

(1) a(iv), b(i), c(ii), d(iii)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(i), c(iii), d(ii)

66. Choose the incorrect statement.

(1) Isopentane and neopentane are chain isomers

(2) Hyperconjugation is observed in neopentyl carbocation

(3) Propanone exhibits keto-enol tautomerism

(4) Propan-1-amine and N-Methylethanamine are functional group isomers.

67. Oxidation state of central carbon atom in carbon suboxide is

(1) Zero

(2) +1

(3) +2

(4) +4

68. Given below are the two statements:
Statement I: Both water gas and producer gas are mixtures in which carbon monoxide is present as one of the components
Statement II: On small scale, pure CO is prepared by dehydration of formic acid with concentrated H_2SO_4 at 373 K.
In the light of above statements, choose correct option.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

69. Given below are the two statements

Statement I: Aniline reacts with concentrated sulphuric acid to form anilinium hydrogensulphate which on heating with sulphuric acid at 453-473 K produces sulphanilic acid as a major product.

Statement II: Direct nitration of aniline at 288 K gives 51% p-nitroaniline, 2% m-nitroaniline and 47% o-nitroaniline.

In the light of above statements, choose the correct answer from the options given below.

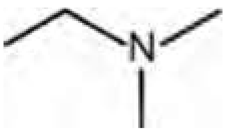
(1) Statement I is correct but statement II is incorrect

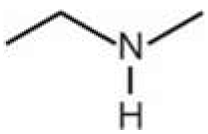
(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

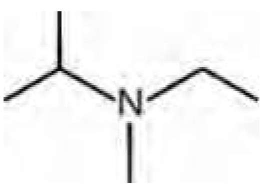
(4) Both statement I and statement II are incorrect

70. Identify the compound that will react with Hinsberg's reagent to give a compound which is insoluble in alkali.

(1) 

(2) 

(3) 

(4) 

71. Match list-I with list-II

	List-I (Octahedral complex)		List-II (Moles of AgCl precipitated per mole of the compound with excess $AgNO_3$)
a.	$CoCl_3 \cdot 4NH_3$	(i)	0
b.	$CoCl_3 \cdot 5NH_3$	(ii)	3
c.	$CoCl_3 \cdot 3NH_3$	(iii)	2
d.	$CoCl_3 \cdot 6NH_3$	(iv)	1

The correct match is

(1) a(iv), b(iii), c(i), d(ii)

(2) a(i), b(iii), c(ii), d(iv)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(iii), b(ii), c(iv), d(i)

72. Consider the following metal carbonyls

a. $[Mn(CO)_6]^+$

b. $[Ni(CO)_4]$

c. $[V(CO)_6]^-$

The correct order of C – O bond length in the metal carbonyls will be

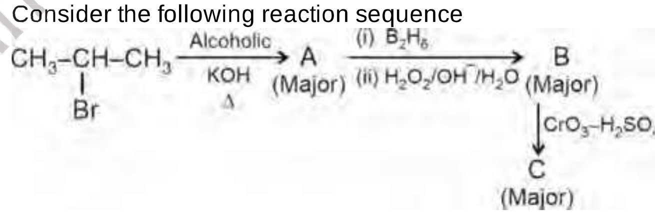
(1) a > b > c

(2) c > b > a

(3) b > a > c

(4) b > c > a

73. Consider the following reaction sequence



Major product C is a/an

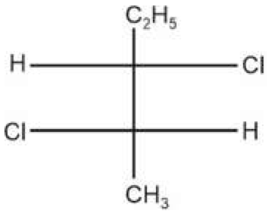
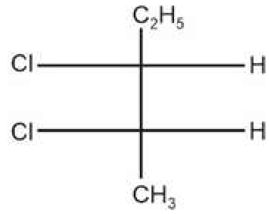
(1) Carboxylic acid

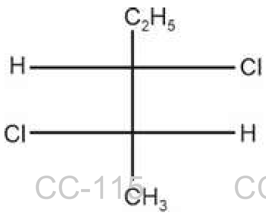
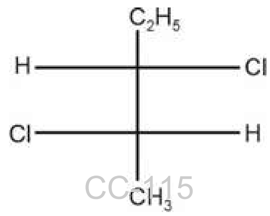
(2) Aldehyde

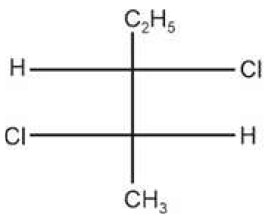
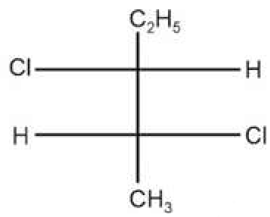
(3) Ketone

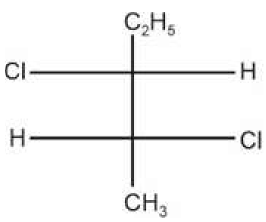
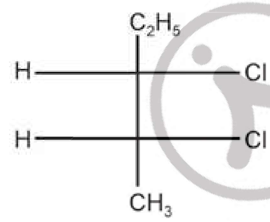
(4) Alkene

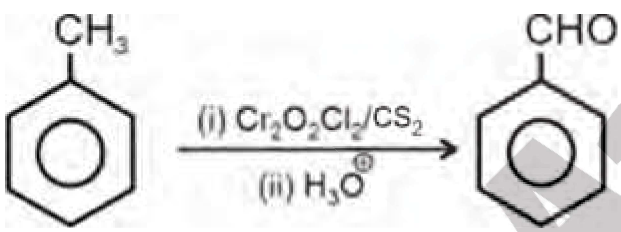
74. Which of the following pairs of compounds are enantiomers to each other?

(1)  and 

(2)  and 

(3)  and 

(4)  and 

75. 

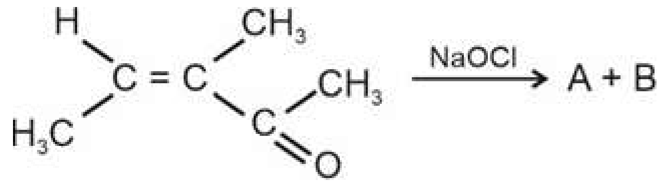
Above reaction is named as

(1) Stephen's reaction

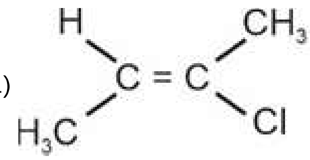
(2) Etard reaction

(3) Gatterman Koch reaction

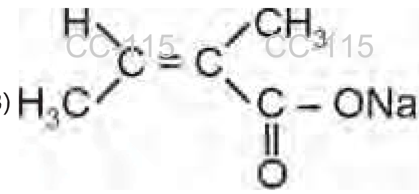
(4) Reimer Tiemann reaction

76. 

Products A and B are

(1)  and CH_3COOH

(2) CH_3COOH and $CH_3 - CH_2 - COCH_2Cl$

(3)  and $CHCl_3$

(4) CH_3COONa and $CHCl_3$

77. Mechanism of a hypothetical reaction $2A_3 \rightarrow 3A_2$ is given below.

(I) $A_3 \rightleftharpoons A_2 + A$ (fast)

(II) $A_3 + A \rightleftharpoons 2A_2$ (slow)

Rate law for reaction will be

(1) $r = k[A_3]^2[A_2]^{-1}$

(2) $r = k[A_3][A_2]$

(3) $r = k[A_3]^2[A_2]$

(4) $r = k[A_3]^2$

78. For a chemical reaction

$H_2O_2\,(aq) + 3I^-\,(aq) + 2H^+\,(aq) \rightarrow$
$2H_2O\,(l) + I_3^-\,(aq)$

If rate of disappearance of I^-(aq) is x mol L^{-1} s^{-1} then rate of appearance of $I_3^-\,(aq)$ will be

(1) 3x

(2) $\frac{x}{3}$

(3) x

(4) $\frac{1}{x}$

79. Match group of cation given in list-I with their group reagent for qualitative analysis given in list-II

	List-I		List-II
a.	Group-I	(i)	NH_4OH+NH_4Cl
b.	Group-II	(ii)	dil. HCl
c.	Group-III	(iii)	H_2S + dil. HCl
d.	Group-IV	(iv)	H_2S+NH_4OH

Correct match among the following is

(1) a(iii), b(ii), c(iv), d(i)

(2) a(iii), b(ii), c(i), d(iv)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(ii), b(iii), c(i), d(iv)

80. Standard molar enthalpy of formation of which pair of substances at 298 K has zero value?

(1) N_2(g) and $S_{rhombic}$(s)

(2) I_2(s) and P_4(g)

(3) $Br_2(l)$ and HF(g)

(4) $C_{diamond}$(s) and Cl_2(g)

81. Given below are the two statements, one is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.
Assertion (A): The freezing point of 0.1 m glucose solution is higher than that of 0.1 m NaCl
Reason (R): Depression in freezing point is inversely proportional to the number of moles of non-volatile solute particles present in the solution.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

82. 10 g of a non-electrolyte solute is dissolved in 114 g of n-octane, it reduces the vapour pressure to 80%. The molar mass (in g mol^{-1}) of the solute is

(1) 20

(2) 40

(3) 80

(4) 60

83. $p\pi$-$d\pi$ bond is absent in

(1) CO_3^{2-}

(2) SO_3

(3) PO_4^{3-}

(4) ClO_4^-

84. In which of the following processes bond order increases.

(1) N_2 changes to N_2^+

(2) C_2 changes to C_2^+

(3) O_2 changes to O_2^-

(4) O_2 changes to O_2^+

85. Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The electrons placed in the antibonding molecular orbital destabilize the molecule.
Reason (R): The mutual repulsion of electrons in antibonding molecular orbital is more than the attraction between the electrons and nuclei, which causes net increase in energy.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

86. Consider the following statements:
Statement I: Both Eu^{2+} ang Gd^{2+} are paramagnetic in nature.
Statement II: $Gd(OH)_3$ is more basic in nature than $Nd(OH)_3$.
In the light of above statements, choose correct option.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

87. Identify the correct order of oxidising power for the following species.

(1) $Cr_2O_7^{2-} < VO_2^+ < MnO_4^-$

(2) $MnO_4^- < Cr_2O_7^{2-} < VO_2^+$

(3) $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$

(4) $MnO_4^- < VO_2^+ < Cr_2O_7^{2-}$

88. The wavelength of light emitted when the electron in a hydrogen atom undergoes transition from an energy level with n = 5 to the energy level with n = 2 is (R_H = Rydberg constant of hydrogen atom)

(1) $\frac{1}{25R_H}$

(2) $\frac{10}{3R_H}$

(3) $\frac{100}{21R_H}$

(4) $\frac{81}{4R_H}$

89. Consider the following statements about the catalyst
a. It lowers the activation energy for the forward and reverse reactions by exactly the same extent.
b. It increases the equilibrium constant value.
c. It does not appear in the balanced chemical equation.
The correct statements are

(1) a and b only

(2) a and c only

(3) b and c only

(4) a, b and c

90. Choose the incorrect match among the following

Physical Quantity		**SI unit**
a. Electric current	(i)	Ampere
b. Amount of substance	(ii)	Kilogram
c. Luminous intensity	(iii)	Candela
d. Thermodynamic temperature	(iv)	Kelvin

Choose the correct option.

(1) a(i)

(2) b(ii)

(3) c(iii)

(4) d(iv)

BIOLOGY

91. Which of the following is **correct** regarding the taxa of different taxonomic categories showing hierarchial arrangement in ascending order?

(1) *Homo* → Hominidae → Mammalia → Primata

(2) *Triticum* → Poaceae → Poales → Monocotyledonae

(3) *Musca* → Muscidae → Insecta → Diptera

(4) *Mangifera* → Anacardiaceae → Dicotyledonae → Sapindales

92. Read the following statements and select the **correct** option.
Statement A: The pigments of euglenoids are identical to those present in higher plants.
Statement B: All protozoans are heterotrophs and live as predators or parasites.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statement A and statement B are correct

(4) Both statement A and statement B are incorrect

93. Sleeping sickness and bovine spongiform encephalopathy are respectively caused by

(1) *Trypanosoma* and prions

(2) *Plasmodium* and viroids

(3) *Entamoeba* and viruses

(4) *Paramoecium* and virusoids

94. Members of the plant family that exhibits the below given floral formula, show the presence of

$$Br \oplus ⚥ \overset{\frown}{P_{(3+3)}\,A_{3+3}}\,\underline{G}_{(3)}$$

(1) Exstipulate leaves with parallel venation

(2) Non-endospermous seeds

(3) Trilocular inferior ovary with many ovules

(4) Valvate aestivation and basal placentation

95. Identify the **incorrect** statement.

(1) The hilum is a scar on the seed coat through which the developing seeds were attached to the fruit.

(2) In coconut, the mesocarp is fibrous.

(3) The outer covering of endosperm in maize seed separates the embryo by proteinaceous layer called aleurone layer.

(4) The plumule and radicle are enclosed in sheaths which are called coleorhiza and coleoptile respectively.

96. Identify the **mismatched** pair.

(1) Solanaceae – Tobacco

(2) Graminae – Wheat

(3) Liliaceae – *Trifolium*

(4) Fabaceae – Muliathi

97. Select the **incorrectly** matched pair.

(1) *Albugo* – Parasitic fungi on mustard

(2) *Rhizopus* – Aseptate and coenocytic mycelium

(3) *Puccinia* – Asexual reproduction is mostly by conidia

(4) *Agaricus* – Karyogamy and meiosis take place in the basidium

98. Which of the following features are associated with prokaryotes but not typically found in eukaryotes?
a. Fimbriae b. Flagella
c. Ribosomes d. Gas vacuoles
e. Cell wall
Select the **correct** answer from the following options

(1) a, d and e only

(2) b, c and d only

(3) a and e only

(4) a and d only

99. Identify the following statements as **true (T)** or **false (F)** and choose the **correct** option.
A. Gas vacuoles are not bound by any unit membrane system.
B. The quasi-fluid nature of lipids in cell membrane enables the lateral movement of proteins within the overall bilayer.
C. Position of future cell plate in plant cells is determined by microtubules.
D. In flagella, the axoneme usually has nine triplets of radially arranged peripheral microtubules and a central hub.

	A	B	C	D
(1)	T	F	F	T
(2)	T	T	T	F
(3)	T	F	T	F
(4)	F	T	F	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

100. What will be the amount of DNA in each meiosis-II product, if the meiocyte contains 20 pg of DNA in G_1 phase?

(1) 40 pg

(2) 20 pg

(3) 10 pg

(4) 60 pg

101. Select the events that are associated with the interphase.
a. Duplication of centrioles.
b. Attachment of spindle fibres to the kinetochores.
c. Exchange of genetic material between homologous chromosomes.
d. Synthesis of tubulin proteins required for forming mitotic apparatus.
e. Chromatin condensation.
The **correct** ones are

(1) a, c, d and e only

(2) a and d only

(3) b and e only

(4) c, d and e only

102. True nucleus is observed in

(1) *E. coli*

(2) Mycoplasma

(3) *Volvox*

(4) *Nostoc*

103. Read the following statements and choose the **incorrect** one.

(1) Crossing over is an enzyme mediated process

(2) In oocytes of some vertebrates, diplotene can last for months or years

(3) The two asters together with spindle fibres and kinetochores form the mitotic apparatus

(4) Telophase is the last phase of karyokinesis in the mitotic division

104. Which of the following tissues generally has isodiametric cells and forms major component within the plant organs?

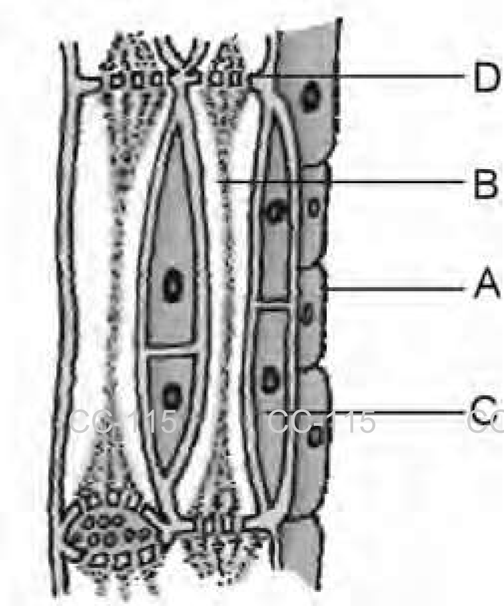
(1) Sclerenchyma

(2) Sclereid

(3) Collenchyma

(4) Parenchyma

105. Examine the figure given below and choose the **correct** option for it.



(1) 'D' represents the structures through which major transport of water and minerals occurs.

(2) 'A' represents the parenchymatous cells that help in maintaining pressure gradient in the element of phloem.

(3) 'B' represents the only dead element of phloem.

(4) 'C' represents the long, narrow cells that are found to be absent in gymnosperms and pteridophytes.

106. Hypothalamus releases a hormone 'X' which stimulates the pituitary gland to synthesise and release 'Y'. Now, 'Y' majorly acts on the Leydig cells and stimulates the synthesis of hormones from gonads.
Identify the **correct** option w.r.t. the primary and direct role of 'Y' in the human body.

(1) Induces ovulation and maintains corpus luteum

(2) Regulates the development of secondary sexual characteristics

(3) Acts on mammary glands and stimulates the formation of alveoli.

(4) Maintains the functions of male sex accessory ducts

107. Select the **incorrect** option w.r.t. colostrum.

(1) Yellowish fluid secreted by the mother during initial days of lactation period.

(2) Has abundant amount of immunoglobulins and nutrients

(3) Provides artificially acquired passive immunity to the baby

(4) Protects the infants from acquiring various infections.

108. Select the **incorrect** statement.

(1) Narcotic analgesics can be misused by athletes.

(2) Opioids such as morphine have a clinical significance.

(3) Deepening of voice is a side-effect of misuse of anabolic steroids in females.

(4) Only the roots of the Cannabis plant are used for obtaining hashish or various other products

109. Benign tumors differ from malignant tumors as

(1) Former remain confined to their original location while latter invade other neighbouring sites

(2) Former grow rapidly while latter grow slowly

(3) Cells in former exhibit contact inhibition while latter do not

(4) Former show neoplastic transformation while latter do not

110. Consider the following statements:
Statement A: Underproduction of steroidal hormones by adrenal medulla alters the carbohydrate metabolism causing acute weakness and fatigue leading to Addison's disease.
Statement B: Pancreas is a composite gland with more than 95 per cent part of it being exocrine in nature.
Select the **correct** option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is incorrect

(4) Only statement B is incorrect

111. Select the **correct** match.

(1)	Restriction enzymes	–	Nucleases that belong to the category of hydrolases
(2)	Agarose compound	–	Natural monomer extracted from sea weeds
(2)	*kan*^{R}	–	Kanamycin resistance encoding gene present in pBR322
(4)	*rop*	–	Codes for the proteins that affect copy number but not involved in the replication of the plasmid

(1) (1)

(2) (2)

(3) (3)

(4) (4)

112. If a plasmid vector has been cut by *Sal I* into a number of fragments then which technique can be employed to check the restriction digestion progression?

(1) ELISA in the presence of chromogenic substrate

(2) Probe hybridization without performing autoradiography

(3) Northern blotting by precipitating the sample in alcohol

(4) Agarose gel electrophoresis

113. Use of PCR can be employed in
a. Archaeogenetics to amplify and extract DNA from fossils to study.
b. Detection and diagnosis of various infectious diseases.
c. DNA forensics for crime investigation
Choose the correct option.

(1) Only a

(2) Only b and c

(3) Only c

(4) a, b and c

114. For making the cell competent to take up the foreign DNA, the bacterial cell is treated with

(1) Na^+

(2) K^+

(3) Cl^-

(4) Ca^{+2}

115. Which of the following is true for a neuron when it is not conducting any impulse, during the resting stage?

(1) Rapid influx of Na^+ occurs to maintain the polarised state

(2) Rapid efflux of negatively charged proteins occurs from axoplasm to ECF

(3) Active transport of ions occurs by the sodium-potassium pump to maintain the electrochemical gradient of the cell

(4) Rapid influx of K^+ through voltage gated channels occurs inside the axonal membrane

116. Read the given assertion (A) and reason (R) and choose the **correct** option.
Assertion (A): During Griffith's experiment, the R-strain bacteria had somehow been transformed by heat-killed S-strain bacteria.
Reason (R): There might be absorption of some transforming principle by rough type bacteria from heat-killed smooth bacteria during Griffith's experiment, that had enabled R-strain to synthesise smooth polysaccharide coat.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

117. During the process of transcription, both the strands of DNA are not copied as
a. They would form two RNA molecules that are complementary to each other.
b. It may lead to prevention of translation of RNA into protein.
c. Same type of proteins will be produced in higher quantity.
d. They lack structural and promoter genes
The **correct** ones are

(1) b and c only

(2) c and d only

(3) a, c and d only

(4) a and b only

118. State true **(T)** or false **(F)** for the given statements and choose the **correct** option
A. One of the applications of human genome project is to correct a number of hereditary diseases produced due to single gene defects.
B. The human genome contains 3164.7 billion nucleotide bases.

	A	B
(1)	T	T
(2)	F	F
(3)	T	F
(4)	F	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

119. Match column I with column II.

	Column I		Column II
a.	Bulliform cells	(i)	Composed of dumb-bell shaped guard cells and tiny pore in the leaves of grasses
b.	Cuticle	(ii)	Usually single layered and present in most plant parts
c.	Stomata	(iii)	Large, empty and colourless cells found in monocots
d.	Epidermis	(iv)	Waxy thick layer prevents water loss

Choose the **correct** option.

(1) a(ii), b(iii), c(iv), d(i)

(2) a(iii), b(iv), c(i), d(ii)

(3) a(i), b(ii), c(iii), d(iv)

(4) a(iv), b(i), c(ii), d(iii)

120. The role of a ribozyme (23S rRNA) during translation is to

(1) Breakdown the proteins into smaller fragments

(2) Synthesize the mRNA

(3) Catalyse the formation of peptide bond

(4) Catalyse aminoacylation of tRNA

121. Select the **incorrect** option w.r.t. addition of an unusual nucleotide, *i.e.*, methyl guanosine triphosphate in primary transcript.

(1) Essential for formation of mRNA-ribosome complex

(2) Capping occurs rapidly after the start of transcription

(3) Occurs at the 3′ end in template dependent manner

(4) Catalysed by guanylyl transferase

122. Which of the following hormones breaks seed and bud dormancy and initiates germination in peanut seeds?

(1) Abscisic acid

(2) Ethylene

(3) Zeatin

(4) Auxin

123. Identify the **incorrect** match from the following

(1) Auxin – NAA

(2) Gibberellins – Indole compound

(3) Ethylene – Gaseous plant hormone

(4) Cytokinin – Present in coconut milk

124. Identify the **incorrect** statement from the following w.r.t. the pollen grains.

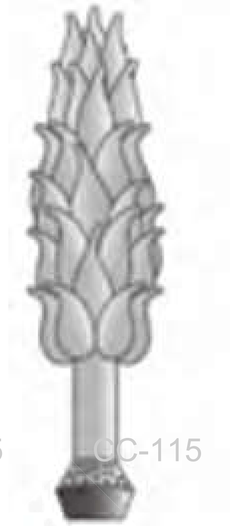
(1) Vegetative cell has a large and irregularly shaped nucleus.

(2) Generative cell is spindle shaped.

(3) Intine exhibits a fascinating array of patterns and designs.

(4) Their viability is highly variable and depends on prevailing temperature and humidity.

125. Observe the figure given below.



The above figure represents

(1) Multicarpellary, syncarpous gynoecium of *Papaver*

(2) Multicarpellary, syncarpous gynoecium of *Michelia*

(3) Multicarpellary, apocarpous gynoecium of *Michelia*

(4) Multicarpellary, apocarpous gynoecium of *Papaver*

126. A breeder wants to perform artificial hybridisation experiments. To ensure that only the desired pollens fall on the stigma, he should make sure that
a. Emasculation of staminate flower is done.
b. Bagging of the emasculated flower is done.
c. Bagging of the female flower in bud stage is done.
d. Bisexual flowers, showing protogyny are not selected for the cross.
The **correct** ones are

(1) b, c and d only

(2) b and c only

(3) a, b and d only

(4) a and c only

127. Phenotypic ratio of the F_2 generation in a Mendelian monohybrid cross in a pea plant is

(1) 1 : 1

(2) 3 : 1

(3) 1 : 2 : 1

(4) 9 : 3 : 3 : 1

128. Which of the given statements is **true** about the recessive traits?

(1) They are not expressed in haploid species

(2) They require two copies of the recessive allele to be expressed in diploid species

(3) They are expressed in the heterozygous condition

(4) They can mask the expression of dominant traits

129. Read the statements given below and choose the **correct** answer.

Statement I: Linked genes always assort independently of one another.

Statement II: The frequency of recombination between gene pairs on the same chromosome can be used to determine their map distance on the chromosome.

(1) Statement I is true but statement II is false

(2) Statement II is true but statement I is false

(3) Both the statements I and II are true

(4) Both the statements I and II are false

130. Match column I with column II.

	Column I		Column II
a.	Sickle-cell anaemia	(i)	Occurs due to the presence of defective blood clotting factor in affected individual
b.	α-Thalassemia	(ii)	Affected individual lacks liver enzyme phenylalanine hydroxylase
c.	Phenylketonuria	(iii)	Defect is caused by transversion mutation of the gene controlling β-chain of haemoglobin
d.	Haemophilia	(iv)	Controlled by two closely linked genes HBA1 and HBA2 on chromosome 16

Choose the **correct** option.

(1) a(iv), b(ii), c(i), d(iii)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(i), b(iii), c(iv), d(ii)

131. Choose the property which is common to both nerve cells and cardiac muscle cells.

(1) Elasticity

(2) Excitability

(3) Extensibility

(4) Contractility

132. If a gene of interest is ligated at *Pst* I restriction site of pBR322, and inserted into natural *E. coli*, then the non-recombinant transformants formed will be

(1) $amp^S tet^R$

(2) $amp^R tet^R$

(3) $amp^S tet^S$

(4) $amp^R tet^S$

133. Complete the analogy by selecting the **correct** option.
Unicellular exocrine gland: Goblet cells :: Multicellular exocrine gland: _________

(1) Thymus gland

(2) Salivary gland

(3) Thyroid gland

(4) Adrenal gland

134. The chemical and physical properties of amino acids are essentially due to

a. Amino group

b. Carboxyl group

c. R functional group

Choose the option with the **correct** set.

(1) a and b only

(2) b and c only

(3) a and c only

(4) a, b and c

135. In *Periplaneta americana*, the head is connected with thorax by a short extension of the _______ known as the neck. Choose the option which fills the blank correctly.

(1) Labrum

(2) Labium

(3) Mesothorax

(4) Prothorax

136. Assertion (A): The valve which guards the opening between right atrium and right ventricle of human heart is called bicuspid valve.

Reason (R): The valve that guards the opening between right atrium and right ventricle is formed by two muscular cusps.

In the light of above statemetns, choose the correct answer from the options given below.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

137. Select the **incorrect** statement.

(1) Disulphide bond is present in insulin and in antibody molecule but it is absent in glycogen.

(2) Aspartic acid and arachidonic acid are considered as biomacromolecules.

(3) On chemical analysis of a living tissue, the acid-soluble pool represents roughly the cytoplasmic composition.

(4) Chargaff's rule is applicable to DNA but not to RNA.

138. In humans, ciliated epithelium is present in the inner surface of bronchioles. This type of epithelium is also present in the structure that

(1) Acts as the site of fertilization

(2) Aids in digestion of food

(3) Facilitates maximum absorption of substances in the renal tubules

(4) Participates in exchange of gases between blood and atmospheric air

139. The type of biomacromolecules whose monomeric unit is nucleotide constitutes what % of the total cellular mass?

(1) 10-15

(2) 5-7

(3) 70-90

(4) 20-30

140. Read the following statements carefully:
(a) In skeletal muscle, contraction is initiated by a signal sent by the CNS *via* a motor neuron.
(b) Repeated activation of muscles can lead to the accumulation of lactic acid due to aerobic breakdown of glycogen.
(c) Muscle fibre is the anatomical unit of muscles.
Choose the **correct** option.

(1) Only (a) and (c) are correct

(2) Only (a) and (b) are correct

(3) (a), (b) and (c) are correct

(4) Only (b) and (c) are correct

141. Match column I with column II.

Column I		**Column II**
(a) Fishes	(i)	Both pulmonary and systemic circulation are present
(b) Amphibians	(ii)	Most of them contain 3-chambered heart
(c) Reptiles	(iii)	Both oxygenated and deoxygenated blood get mixed up in a single ventricle
(d) Birds	(iv)	Heart pumps out deoxygenated blood

Choose the **correct** option.

(1) a(iv), b(iii), c(i), d(ii)

(2) a(iv), b(ii), c(i), d(iii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(i), b(ii), c(iii), d(iv)

142. The formed element which is majorly involved in allergic reactions and in providing protection against parasitic infections is

(1) Monocyte

(2) Neutrophil

(3) Eosinophil

(4) Erythrocyte

143. Select the unpaired cranial bone present in humans.

(1) Ethmoid

(2) Parietal

(3) Temporal

(4) Zygomatic

144. Choose the **incorrect** option w.r.t. the total number of different bones that constitute the adult human skeleton.

(1) Cervical vertebrae – 7

(2) Scapula – 2

(3) Ribs – 24

(4) Pelvic girdle – 4

145. Which of the following organisms can maintain a constant body temperature?
a. *Calotes*
b. *Struthio*
c. *Neophron*
d. *Elephas*
Select the **correct** option.

(1) a and b

(2) a, c and d

(3) a and d

(4) b, c and d

146. The Neanderthal man had a cranial capacity of

(1) 900cc

(2) 1400cc

(3) 650cc

(4) 800cc

147. The key concepts of Darwinian theory of evolution are

(1) Branching descent and mutations

(2) Natural selection and branching descent

(3) Non-directional variations

(4) Large and random variations

148. If we consider the chronological order of human evolution from early to most recent period, then

(1) *Australopithecines* will be placed after *Homo habilis*

(2) *Homo erectus* will be placed before the Neanderthal man

(3) *Australopithecines* will be placed earlier than *Dryopithecus*

(4) *Homo erectus* will be placed earlier than *Homo habilis*

149. Read the following statements w.r.t. the Henle's loop in humans:
(A) It plays a significant role in the maintenance of low osmolarity of the medullary interstitial fluid.
(B) Its descending limb is permeable to electrolytes but impermeable to water.
(C) Reabsorption is maximum in its ascending limb.
(D) Its ascending limb allows transport of electrolytes actively or passively.
Which of the above given statements are **not** true?

(1) (A) and (B) only
(2) (C) and (D) only
(3) (A), (B) and (D) only
(4) (A), (B) and (C) only

150. Observe the matches given below and select the **correctly** matched option.

(1)	Marsupial rat and tiger cat	–	Convergent evolution
(2)	Eyes of *Octopus* and that of dogs	–	Divergent evolution
(3)	Banded anteater and sugar glider	–	Adaptive radiation
(4)	Mouse and Lemur	–	Convergent evolution

(1) (1)
(2) (2)
(3) (3)
(4) (4)

151. In humans, counter current mechanism helps in concentrating the filtrate. This mechanism involves the direct participation of
(a) PCT
(b) Henle's loop
(c) DCT
(d) Vasa recta
Choose the **correct** option.

(1) (a) and (b)
(2) (c) and (d)
(3) (a) and (c)
(4) (b) and (d)

152. A person went for an event to Hong Kong *via* flight, where he came in contact with a person who was diagnosed with COVID few days later. Now, upon his arrival to India, he was found to be asymptomatic but was still put in isolation to prevent the spread of the virus. Which of the following techniques is used to early diagnose whether he is infected with corona virus or not?

(1) Serum analysis
(2) Biopsy
(3) RT-PCR
(4) Urine analysis

153. Exclusively marine animals with worm-like body and stomochord have

(1) Closed circulatory system
(2) Open circulatory system
(3) Incomplete digestive system
(4) Endoskeleton of calcareous ossicles

154. A novel mechanism of RNA interference inhibits

(1) Transcription of specific mRNA of the host plant
(2) DNA replication within the host plant
(3) Translation of specific genes in prokaryotes
(4) Translation of specific mRNA in eukaryotes

155. A pregnant female went for amniocentesis in a hospital after recommendation from the consulting doctor due to history of genetic disorders in the family. Her reports suggested that the foetus has mutated ADA gene. Which of the following is likely to be the best approach for the permanent cure of this disease?

(1) Bone marrow transplantation after the symptoms are apparent in the baby
(2) Enzyme replacement therapy during the embryonic stage
(3) Correction of the gene by the addition of normal ADA gene after the birth
(4) Introduction of normal ADA genes into cells at early embryonic stages.

156. Read the assertion (A) and reason (R) given below and choose the **correct** option.
Assertion (A): In gymnosperms, seeds are not enclosed within the fruit wall.
Reason (R): Ovules become transformed into seeds as the female gametophyte is retained within the ovule on the sporophyte plant of gymnosperm.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

157. In pteridophytes, the main plant body

(1) Is differentiated into true root, stem and leaves
(2) Is haploid and produces gametes
(3) Bears antheridia and archegonia as female and male sex organs, respectively
(4) Is multicellular well differentiated prothallus

158. Select the **mismatched** pair.

(1) Cyclosporin A – Immunosuppressive agent

(2) Statins – Blood-cholesterol lowering agent

(3) Streptokinase – Product of *Staphylococcus*

(4) Lipases – Helpful in removing oily stains

159. Read the following statements (A-D)

A. The size of the population tells us a lot about its status in the habitat.

B. The density of a population in a given habitat during a given period, fluctuates due to changes only in natality and emigration.

C. A few organisms are eurythermal, but a vast majority of them are stenothermal.

D. The intrinsic rate of natural increase (r) is a measure of the inherent potential of a population to grow.

The **correct** statements are

(1) Only A, B and D

(2) Only A, C and D

(3) Only C and D

(4) Only A and B

160. In the biosphere, heterogeneity can be seen at multiple levels like,

a. Organisms

b. Biomes

c. Species

d. Populations

e. Macromolecules

The **correct** ones are

(1) Only a and c

(2) Only c and d

(3) All a, b, c, d and e

(4) Only b and d

161. Which of the following regions has nearly 1400 species of birds?

(1) New York

(2) Colombia

(3) Greenland

(4) Alaska

162. Members of Phaeophyceae vary in colour from olive green to various shades of brown depending upon the amount of the pigment

(1) Carotenoid

(2) Fucoxanthin

(3) Chlorophyll *a*

(4) Chlorophyll *c*

163. Physical removal of particles from the sewage through filtration and sedimentation is carried out in which stage of sewage treatment?

(1) Biological treatment

(2) Primary treatment

(3) Secondary treatment

(4) Tertiary treatment

164. Which of the following options exemplifies the population interaction where each interacting species is benefitted but can live equally well without association?

(1) Sea anemone attached to the body of hermit crab

(2) Barnacles growing on the back of a whale

(3) Clown fish that lives among sea anemone

(4) *Cuscuta* on hedge plants.

165. Read the following statements.

a. In the chloroplast, the primary acceptor of electron is located towards the inner side of the thylakoid membrane.

b. During Calvin cycle, reduction step involves utilisation of one molecule of ATP and NADPH each, per molecule of CO_2 fixed.

c. The enzyme in *Sorghum* plant which catalyses CO_2 fixation is PEP carboxylase.

Choose the **incorrect** one(s).

(1) b and c

(2) a and b

(3) b only

(4) a and c

166. Which among the following is **not** a product of light reaction of photosynthesis?

(1) NADPH

(2) ATP

(3) Oxygen

(4) $FADH_2$

167. Oxidative decarboxylation reaction in mitochondria is catalysed by

(1) Lactate dehydrogenase

(2) Pyruvate kinase

(3) Pyruvate dehydrogenase

(4) Alcohol dehydrogenase

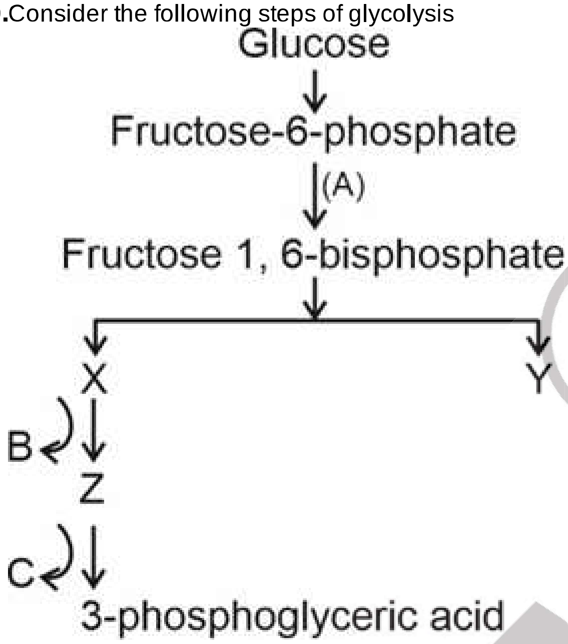
168. Read the given statements (A-E).

A. Earthworm breaks down detritus into smaller particles known as anabolism.
B. The humus is further degraded by some microbes and release only organic nutrients called mineralization.
C. Water soluble inorganic nutrients go into the soil and get precipitated by a process called leaching.
D. Detritus food chain begins with living organism.
E. Rate of decomposition is not controlled by climatic factors.

How many of the above statements are **incorrect**?

(1) Two
(2) Five
(3) Three
(4) Four

169. Consider the following steps of glycolysis



Choose the **correct** option for label A, B and C.

	A	B	C
(1)	Enolase	NADH + H^+	ATP
(2)	Phosphofructokinase	$FADH_2$	NADH
(3)	Phosphohexoisomerase	ATP	ATP
(4)	Phosphofructokinase	NADH+H^+	ATP

(1) (1)
(2) (2)
(3) (3)
(4) (4)

170. The amount of biomass or organic matter produced per unit area over a period of time by plants during photosynthesis is called

(1) Productivity
(2) Primary production
(3) Gross primary productivity
(4) Net primary productivity

171. Assertion (A): *Fasciola* is an endoparasite that contains both hooks and suckers as adaptative structures for survival in the host body.
Reason (R): Liver fluke uses both hooks and suckers to attach to the body of host.
In the light of above statements, choose the **correct** option.

(1) Both (A) and (R) are true and (R) explains (A) correctly.
(2) Both (A) and (R) are true but (R) does not explain (A) correctly.
(3) (A) is true, (R) is false.
(4) Both (A) and (R) are false.

172. In humans, semen consists of

(1) Seminal plasma only
(2) Sperms only
(3) Seminal fluid and prostatic fluid only
(4) Seminal plasma and sperms

173. How many secondary spermatocytes and primary oocytes are required to form 200 million spermatozoa and 200 million secondary oocytes respectively?

(1) 100 million; 200 million
(2) 100 million; 100 million
(3) 200 million; 200 million
(4) 100 million; 400 million

174. Choose the **correct** match w.r.t. humans.

(1) Oogenesis	–	Initiated before the birth of females
(2) Acrosome	–	Filled with hormones
(3) Antrum	–	A cavity present within primary follicles
(4) First polar body	–	Larger than secondary oocyte in size

(1) (1)
(2) (2)
(3) (3)
(4) (4)

175. Which of the following characteristics is common in both frogs and cockroaches?

(1) Body is differentiated into head, thorax, neck and abdomen

(2) Gaseous exchange takes place in the region of tracheoles

(3) Presence of red blood cells

(4) Oviparity

176. Under normal physiological conditions, a person can inspire or expire ‘Y’ mL of air in a minute. This can be represented as

(1) IRV + ERV

(2) VC/TV

(3) Tidal volume × Breathing rate

(4) Vital capacity/Breathing rate

177. Read the following statements w.r.t. humans:

Statement A: We cannot directly alter the pulmonary volume.

Statement B: The anatomical setup of lungs in thorax is such that any change in the volume of the thoracic cavity will be reflected in the pulmonary cavity.

Statement C: Volume of thoracic cavity increases and intrapulmonary pressure decreases during normal expiration.

Choose the **correct** option.

(1) All the statements are incorrect

(2) Only statement A is correct

(3) All the statements are correct

(4) Only statement C is incorrect

178. In frog, the structure which is present in kidneys and is responsible for passage of sperms is

(1) Urinogenital duct

(2) Ureter

(3) Bidder's canal

(4) Cloaca

179. Steroidal oral contraceptive pills which contain only progesterone

(1) Can be taken twice a day

(2) Are once a week pill and can be skipped

(3) Inhibit ovulation and implantation

(4) Never alter the quality of cervical mucus

180. Upon vasectomy in human males, the recently formed sperms from seminiferous tubules cannot reach

(1) Vasa efferentia

(2) Epididymis

(3) Ejaculatory duct

(4) Rete testes